PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
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Department of Computer Science and Engineering

UE24MA241B - Linear Algebra and its Applications

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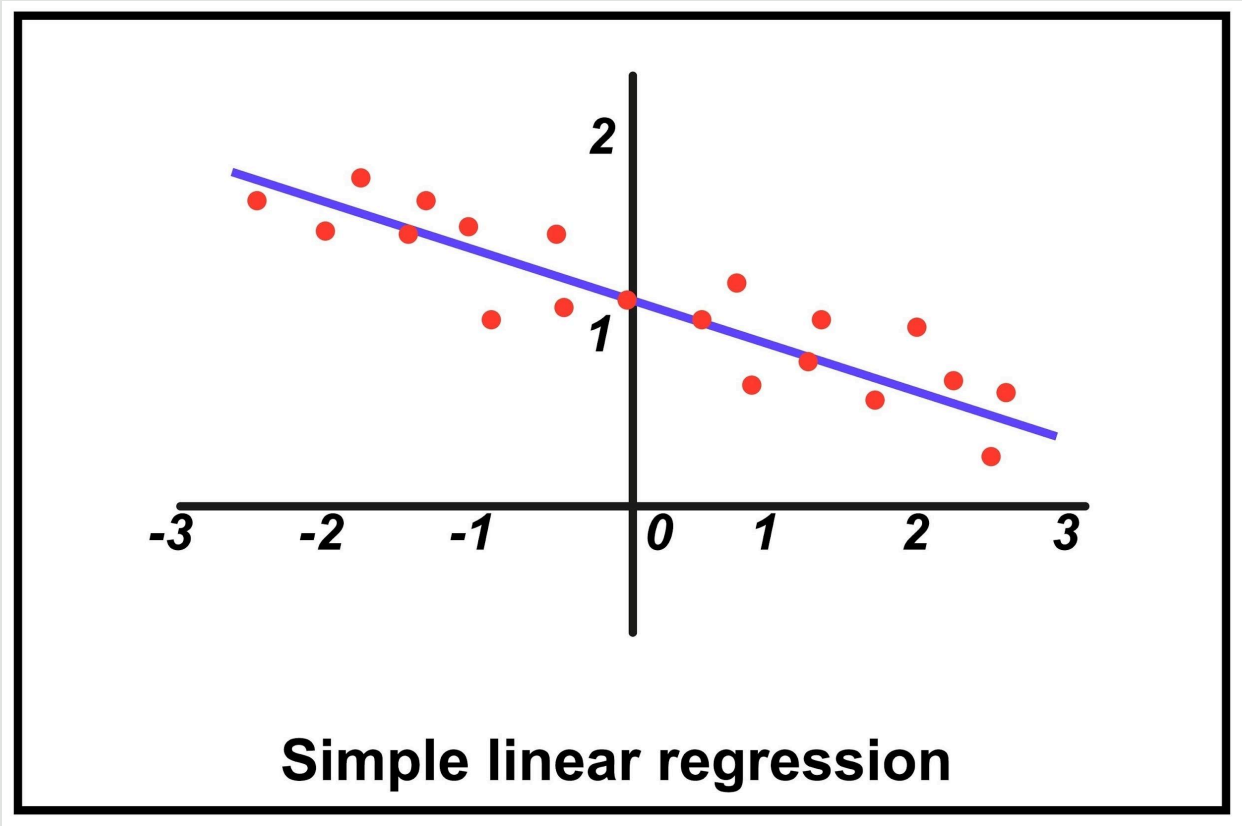
UNIT 3 LINEAR ALGEBRA LIBRARY FUNCTIONS APPENDIX

Gram-Schmidt & Signal Processing (Program 1)

- **np.array()**: Used to define signal vectors v_1 and v_2 as numerical arrays.
- **np.dot(a, b)**: Calculates the dot product (scalar product) of two vectors. In this program, it is used to find the projection of one vector onto another to determine how much of the signals overlap.
- **np.linalg.norm(x)**: Computes the vector L_2 norm (magnitude). This is used during the **Normalization** step to ensure basis vectors have a length of 1.

Least Squares Temperature Prediction (Program 2)

- **np.vstack(tup)**: Stacks arrays vertically. Here, it combines the days array with an array of ones to create the design matrix A.
- **.T (Transpose)**: Flips the matrix over its diagonal. This is applied to the stacked arrays to ensure matrix A has the correct dimensions (rows for observations, columns for variables) for the linear system.
- **np.linalg.lstsq(a, b)**: The core function for the **Least Squares** method. It returns the values for slope (m) and intercept (c) that minimize the squared Euclidean 2-norm $|b - Ax|^2$, providing the best-fit line for the temperature data.



Mechanical Vibrations (Program 3)

- `np.linalg.eig(a)`: Computes the **eigenvalues** and right **eigenvectors** of a square matrix.
    - The **eigenvalues** represent the natural frequencies of the spring-mass system.
    - The **eigenvectors** represent the mode shapes, showing the physical pattern of the vibration.

Google PageRank (Program 4)

- `A.sum(axis=0)`: Sums the elements of the adjacency matrix along the columns. This is used to determine how many outbound links each web page has.
- `np.argmax(a)`: Returns the indices of the maximum values along an axis. In PageRank, it identifies which eigenvalue is the "dominant" one (closest to 1).
- `np.real(val)`: Extracts the real part of a complex number. Since transition matrices can sometimes yield complex eigenvalues/vectors, this ensures the PageRank scores are real-valued.

- **`np.argsort(a)`**: Performs an indirect sort, returning the indices that would sort the array. It is used here with a negative sign (`-pagerank`) to rank the web pages from highest to lowest score.

Image Compression / PCA (Program 5)

- **`np.mean(a, axis=0)`**: Calculates the arithmetic mean. In **Principal Component Analysis (PCA)**, this is used to "center" the data by subtracting the average pixel value from each column.
- **`np.cov(m)`**: Estimates the **covariance matrix** of the data. This matrix describes how different pixels or dimensions in the image vary together.
- **`@` (Matrix Multiplication)**: Used to project the centered image data onto the top k eigenvectors (Principal Components) and then reconstruct it. This achieves compression by ignoring vectors with low eigenvalues.